

**CSE7101- Capstone Project
Review-1**

PROJECT TITLE: PSCS-17 Multilingual Smart Chatbot for PGRKAM

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Name of the Program: Capstone Project

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Name of the Program Project Coordinator: Dr. Jayavadivel Ravi

Name of the School Project Coordinators: Dr. Sampath A K , Dr. Geetha A

Problem Statement Number: 17

Organization: Government OF Punjab

Category (Hardware / Software / Both) : Software

Problem Description: The Punjab Ghar Ghar Rozgar and Karobar Mission (PGRKAM) offers diverse employment and career services through its portal and mobile app, but users struggle to navigate multiple modules to find relevant information. A smart AI-driven chatbot can address this gap by providing instant answers, personalized recommendations, and seamless navigation, improving user experience and ensuring quick access to job and skill development opportunities.

PROBLEM STATEMENT

- The Punjab Ghar Ghar Rozgar and Karobar Mission (PGRKAM) portal and mobile application offer a wide range of employment-related services, including job listings, skill development programs, self-employment opportunities, foreign study counseling, and more.
- However, the absence of an intelligent navigation or query-handling mechanism forces users to manually browse multiple modules, consuming time and effort. There is a need for a smart, multilingual, voice-enabled chatbot that can guide users directly to relevant resources, answer queries, and provide personalized job recommendations.

Objectives

- Develop an AI-driven chatbot using a Large Language Model (LLM) such as GPT-3.5.
- Enable multilingual support (Punjabi, Hindi, English) for both text and voice queries.
- Provide direct answers or navigation guidance for job search, skill development, and counseling-related queries.
- Recommend jobs based on candidate preferences and stored history.

Background and Related Work

- LLM Chatbots: Systems like ChatGPT and Bard use transformer-based architectures to provide natural language understanding and generation.
- Multilingual AI Systems: Research on multilingual NLP models like mBERT and XLM-R has made it possible to process multiple languages with a single model.
- Government Employment Portals: Many existing portals lack conversational interfaces and require manual navigation, which is time-consuming.
- Speech-to-Text & Text-to-Speech: APIs such as Google Speech API and Web Speech API make voice input and output possible in browsers.

Analysis of Problem Statement

Technology Stack Components:

Frontend

React.js – For building an interactive, responsive, and user-friendly web interface.

HTML5, CSS3, JavaScript (ES6) – Core web technologies for layout, styling, and client-side logic.

Express.js – Web framework for building REST APIs and managing chatbot request/response cycles.

AI/Chatbot Engine

OpenAI GPT-3.5 / GPT-4 API – For natural language understanding and intelligent query handling.

Voice & Multilingual Support

Web Speech API – For speech-to-text (voice input) and text-to-speech (audio responses).

Google Cloud Translation API / Indic NLP Library – For multilingual query processing in Punjabi, Hindi, and English.

Database

MongoDB / PostgreSQL – For storing user profiles, chat history, and job preference data.

Github Link

Github Link : <https://github.com/Nidhi-hb>

available files : ppt , frontend , b ackend files ...

Analysis of Problem Statement (contd...)

SOFTWARE REQUIREMENTS : React.js (Latest Version)

HTML5, CSS3, JavaScript (ES6)

Bootstrap / Tailwind CSS

Node js, Express js

Open API (web speech)

Google Cloud Translation API / Indic NLP LIB

Analysis of Problem Statement (contd...)

Mongo DB For Backend Storing Data

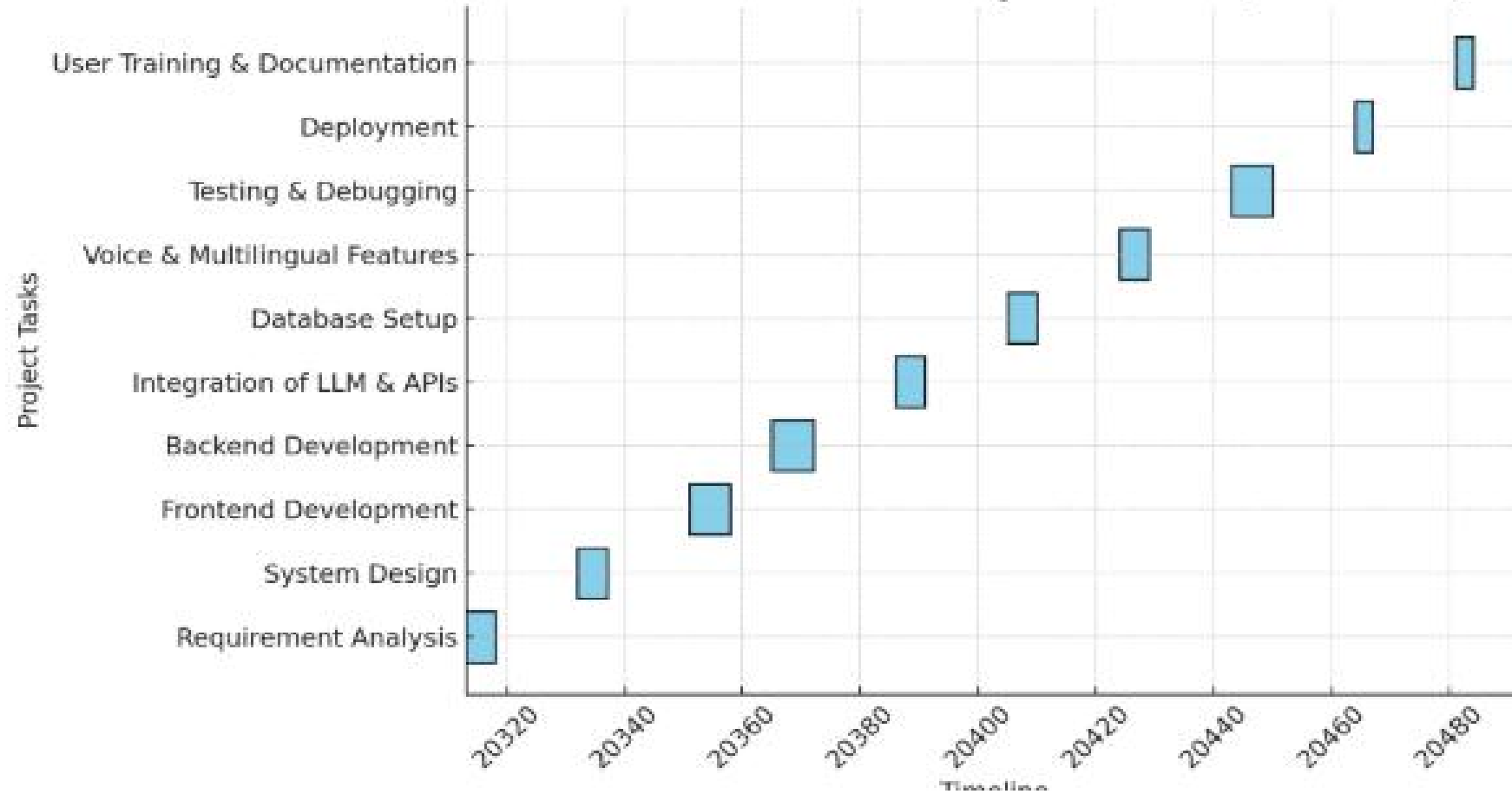
Visual Studio Code (VS Code)

Git & GitHub (Version Control)

Postman (API Testing)

Docker (Optional for Deployment)

PGKKAM Smart Chatbot Project Timeline (Gantt Chart)



References (IEEE Paper format)

Vaswani, A., et al. (2017). "Attention is All You Need." Advances in Neural Information Processing Systems.

Devlin, J., et al. (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." NAACL-HLT.

OpenAI. (2023). GPT-3.5 Turbo Model Documentation.

Punjab Ghar Ghar Rozgar and Karobar Mission (PGRKAM) Official Website.
<https://www.pgrkam.com>

Google Web Speech API Documentation.



Thank
You!



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